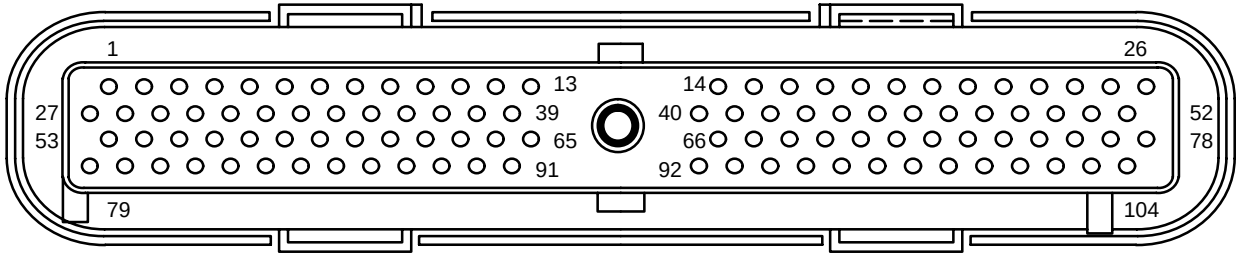
C175

12A581

A147
Powertrain Control
Module (PCM)
(12A650)



F104001

Pin	Circuit	Circuit function
1	1025 (OG/YE)	Coil On Plug (COP) 6 (12029), control
2	–	not used
3	570 (BK/WH)	Ground
4	–	not used
5	–	not used
6	237 (OG/YE)	Shift solenoid A
7	–	not used
8	–	not used
9	–	not used
10	–	not used
11	315 (VT/OG)	Shift solenoid B
12	911 (WH/LG)	Transmission Control Indicator Lamp (TCIL)
13	107 (VT)	Module programming signal
14	–	not used
15	915 (PK/LB)	SCP Bus –
16	914 (TN/OG)	SCP Bus +
17	–	not used
18	–	not used
19	637 (DB/WH)	Air suspension module (3C142), to, Powertrain Control Module (PCM) (12A650)
20	924 (BN/OG)	Powertrain Control Module (PCM) (12A650), to transmission coast clutch solenoid

Pin	Circuit	Circuit function
21	349 (DB)	Crankshaft position sensor (6C315) +
22	350 (GY)	Crankshaft position sensor (6C315) –
23	–	not used
24	–	not used
25	567 (LB/YE)	Power ground
26	1024 (LG/WH)	Coil On Plug (COP) 1 (12029), control
27	1021 (LG/YE)	Coil On Plug (COP) 5 (12029), control
28	–	not used
29	224 (TN/WH)	PCM, to transmission overdrive cancel switch
30	–	not used
31	–	not used
32	311 (DG/VT)	Knock sensor (12A699) –
33	–	not used
34	1144 (YE/BK)	Digital Transmission Range (DTR) sensor, TR1
35	392 (RD/LG)	Heated Oxygen Sensor (HO2S) #12 (9G444)
36	968 (TN/LB)	Mass Air Flow (MAF) sensor (12B579), signal, return
37	923 (OG/BK)	Transmission Fluid Temperature (TFT) sensor, input
38	354 (LG/RD)	Engine Coolant Temperature (ECT) sensor (12A648), input
39	743 (GY)	Intake Air Temperature (IAT) sensor (12A697), input
40	238 (DG/YE)	Fuel pump relay, switched power, output
41	347 (BK/YE)	A/C clutch relay, switched output
42	–	not used

Pin	Circuit	Circuit function
43	71 (OG/LG)	Powertrain Control Module (PCM) (12A650) – Data output link
44	–	not used
45	–	not used
46	367 (BN)	Intake Manifold Runner Control (IMRC) module
47	360 (BN/PK)	EGR Vacuum Regulator (EVR), control
48	–	not used
49	1145 (LB/BK)	Digital Transmission Range (DTR) sensor, TR2
50	1143 (WH/BK)	Digital Transmission Range (DTR) sensor, TR4
51	570 (BK/WH)	Ground
52	1028 (WH/PK)	Coil On Plug (COP) 3 (12029), control
53	1030 (DG/VT)	Coil On Plug (COP) 4 (12029), control
54	480 (VT/YE)	Torque Converter Clutch (TCC) solenoid
55	729 (RD/WH)	Voltage supplied at all times (overload protected)
56	191 (LG/BK)	Vapor management valve
57	310 (YE/RD)	Knock sensor (12A699) +
58	–	not used
59	970 (DG/WH)	Turbine shaft speed sensor (7M101)
60	74 (GY/LB)	Heated Oxygen Sensor (HO2S) #11 (9F472)
61	393 (VT/LG)	Heated Oxygen Sensor (HO2S) #22 (9G444), input
62	791 (RD/PK)	Fuel tank pressure transducer sensor (9C968)
63	–	not used
64	199 (LB/YE)	Digital Transmission Range (DTR) sensor, to PCM
65	352 (BN/LG)	Differential Pressure Feedback EGR (DPFE) sensor, input
66	1102 (YE/LG)	Cylinder–head temperature sensor (6G004), to, Powertrain Control Module (PCM) (12A650)
67	91 (VT/WH)	Canister vent
68	679 (GY/BK)	vehicle speed signal
69	331 (PK/YE)	A/C clutch relay, control
70	–	not used
71	361 (RD)	Voltage supplied in Start and Run (overload protected)
72	561 (TN/RD)	Injector 7, control
73	559 (TN/BK)	Injector 5, control

Pin	Circuit	Circuit function
74	557 (BN/YE)	Injector 3, control
75	555 (TN)	Injector 1, control
76	570 (BK/WH)	Ground
77	–	not used
78	1027 (PK/LB)	Coil On Plug (COP) 7 (12029), control
79	1029 (WH/RD)	Coil On Plug (COP) 8 (12029), control
80	926 (LB/OG)	Fuel pump relay, control
81	925 (WH/YE)	to Electronic Pressure Control (EPC) solenoid
82	–	not used
83	264 (WH/LB)	Idle air control valve (IAC) (9F715), control
84	136 (DB/YE)	Powertrain Control Module (PCM) (12A650) – Output Shaft Speed (OSS) sensor (7M101)
85	795 (DG)	Camshaft position sensor (6B288), to, Powertrain Control Module (PCM) (12A650)
86	–	not used
87	94 (RD/BK)	Heated Oxygen Sensor (HO2S) #21 (9F472)
88	967 (LB/RD)	Mass Air Flow (MAF) sensor (12B579)
89	355 (GY/WH)	Throttle Position Sensor (TPS) (9B989), input
90	351 (BN/WH)	Reference, voltage
91	359 (GY/RD)	signal, return
92	810 (RD/LG)	Brake pedal position switch (13480), input
93	387 (RD/WH)	Heated Oxygen Sensor (HO2S) #11 (9F472), heater
94	388 (YE/LB)	Heated Oxygen Sensor (HO2S) #21 (9F472), heater
95	389 (WH/BK)	Heated Oxygen Sensor (HO2S) #12 (9G444), heater
96	390 (TN/YE)	Heated Oxygen Sensor (HO2S) #22 (9G444), heater
97	361 (RD)	Voltage supplied in Start and Run (overload protected)
98	562 (LB)	Injector 8, control
99	560 (LG/OG)	Injector 6, control
100	558 (BN/LB)	Injector 4, control
101	556 (WH)	Injector 2, control
102	–	not used
103	570 (BK/WH)	Ground
104	1026 (PK/WH)	Coil On Plug (COP) 2 (12029), control